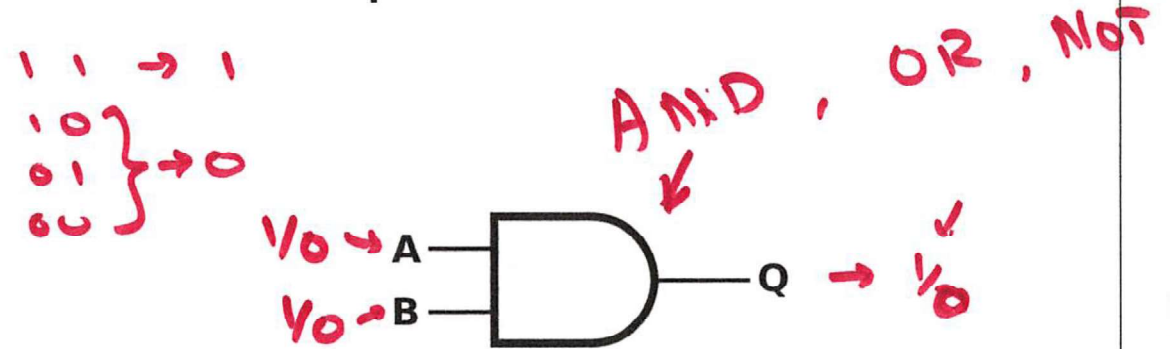
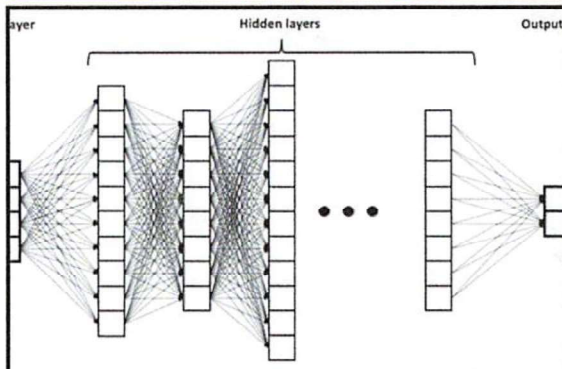
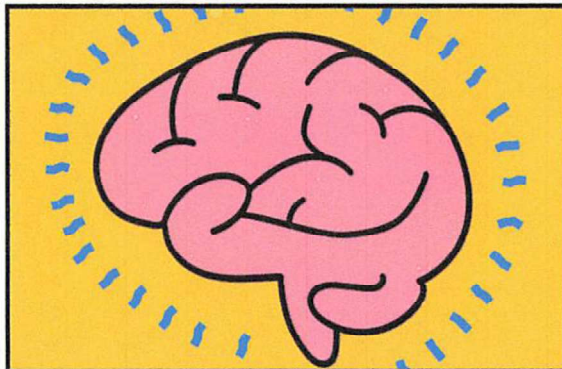


Introduction to Artificial Neural Networks

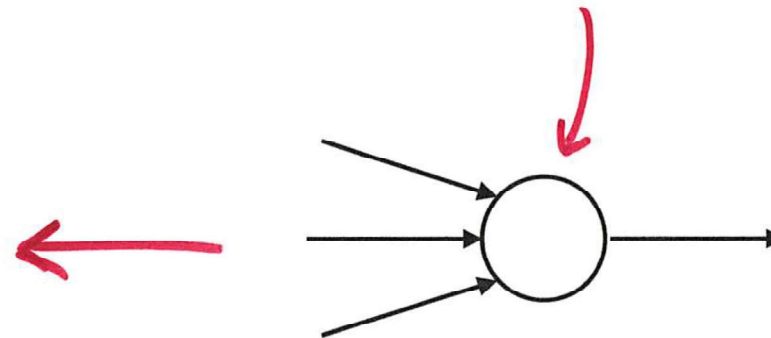
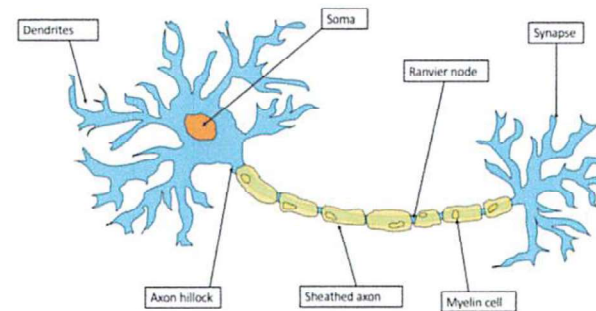
Agenda

- Introduction
- Perceptron
- Multi Layer Perceptron
- An Example - MNIST
- Training
 - Gradient Descent
 - Local vs Global
 - Back Propagation
- Other
 - Activation functions $\sigma =$
 - Loss functions $L =$

Building towards a complex task!



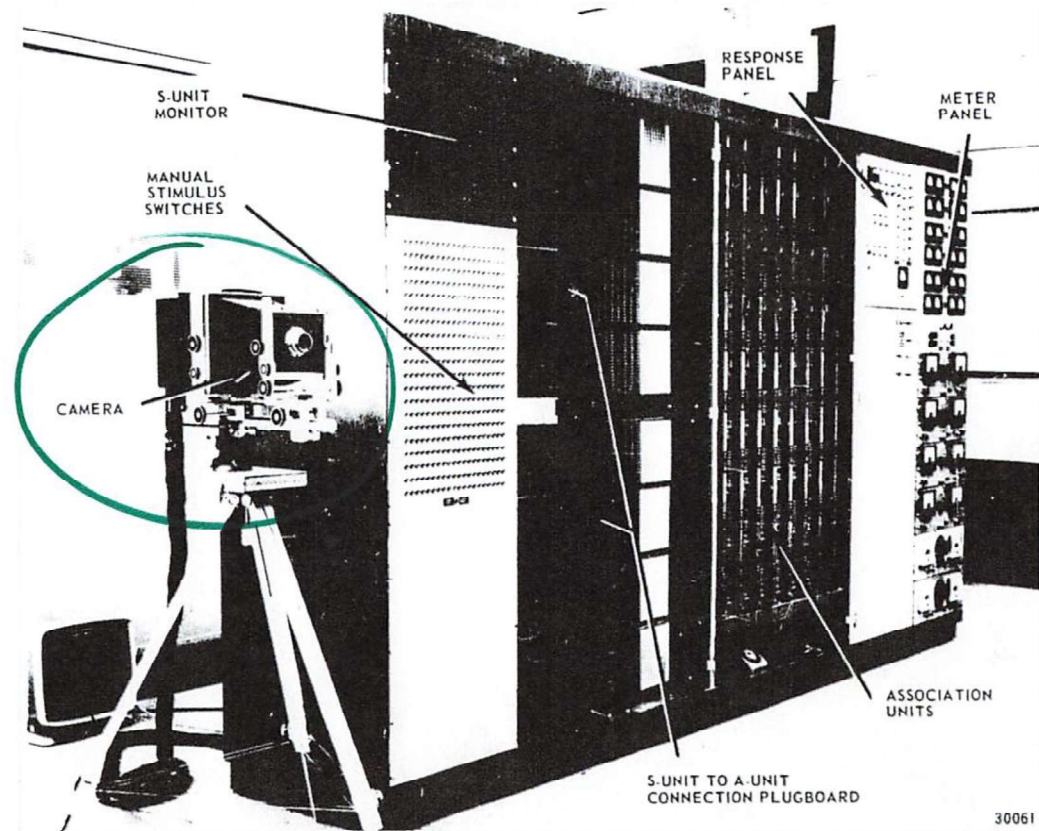
86 B



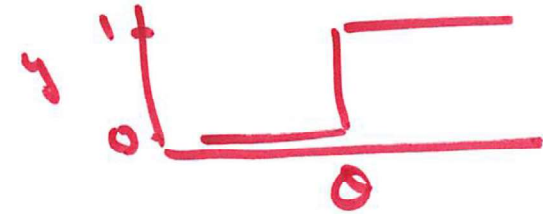
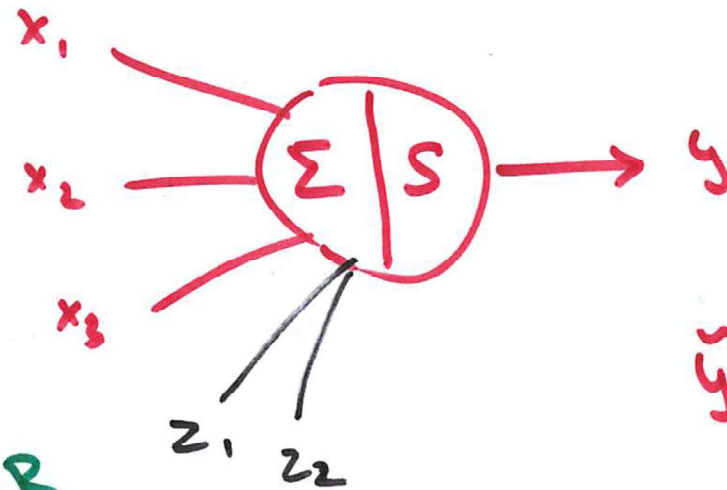
History

- McCulloch-Pitts Neuron, 1940
- "the embryo of an electronic computer that [the Navy] expects will be able to walk talk, see, write reproduce itself and be conscious of its existence." - Rosenblatt, 1958

MARK I
~~PERCEPTRON~~
perceptron
machine



MIP



$$\tilde{y} = \begin{cases} 1 & \text{if } \sum x_i \geq 0 \\ 0 & \text{if } \sum x_i < 0 \end{cases}$$

AND, OR

→ $0 = 3 \rightarrow \text{'and'}$
→ $0 = 1 \rightarrow \text{'or'}$

$$y = \begin{cases} \tilde{y} & \text{only if all } z_i = 0 \\ 0 & \text{if any } z_i = 1 \end{cases}$$

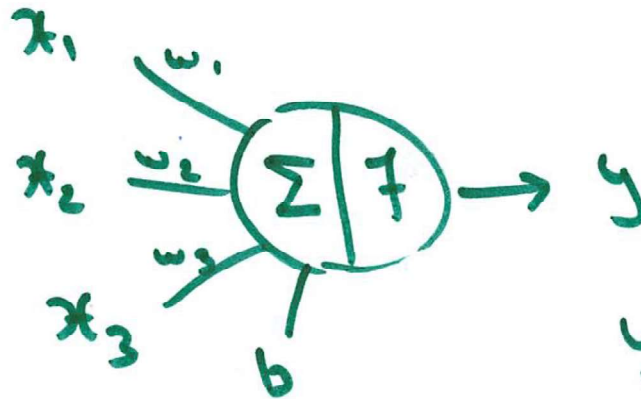
NOT

Perceptron!

$$y = f(\sum x_i w_i)$$

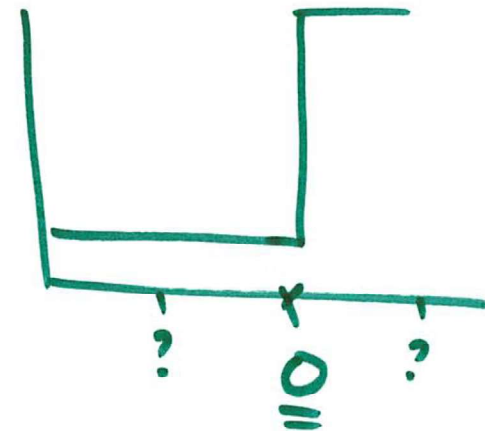
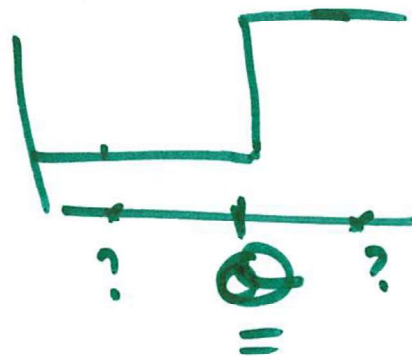
- What does it do?

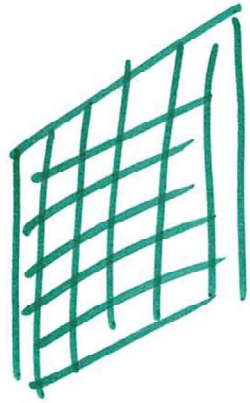
1958



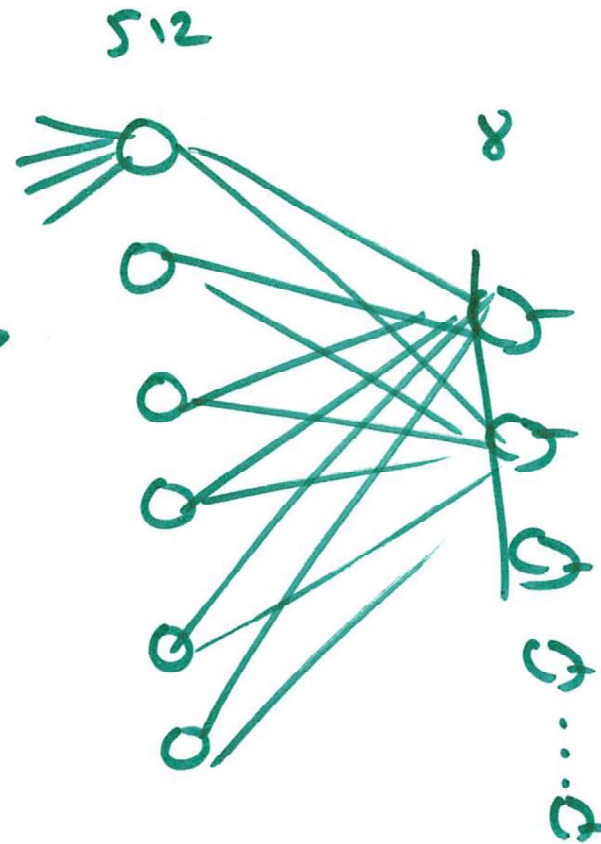
$$y = f(\sum x_i w_i + \underline{b})$$

- Bias

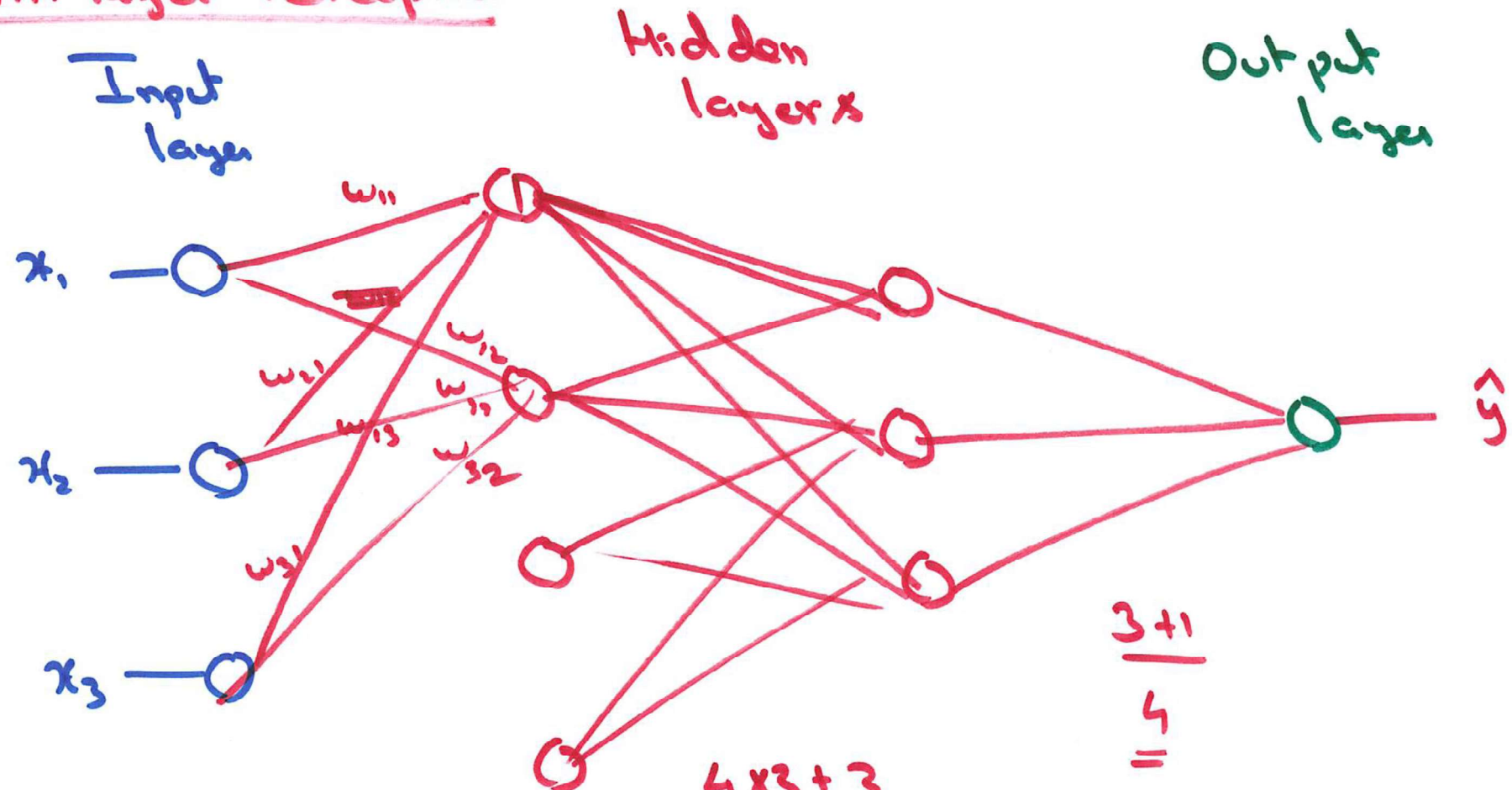




20x20 →



Multi layer Perceptron



$$\underline{3 \times 4 + 4}$$

↓

$$\underline{16}$$

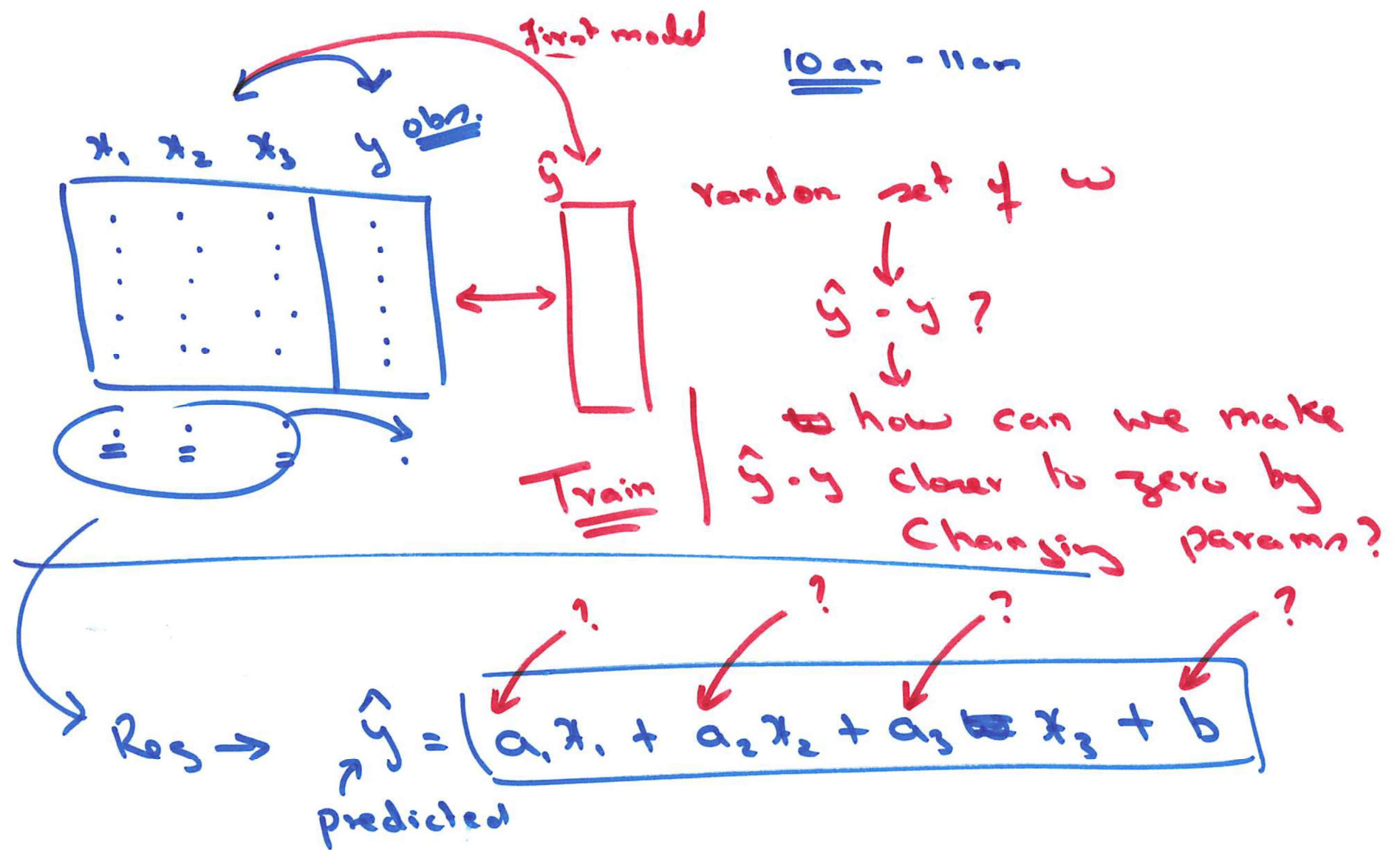
$$\underline{4 \times 3 + 3}$$

↓

$$\underline{15}$$

$$\frac{3+1}{4} =$$

↪ 35 parameters



An Example

⁵ ⁰ ⁴ ¹ ⁹ ² ¹ ³ ¹ ⁴
5 0 4 1 9 2 1 3 1 4

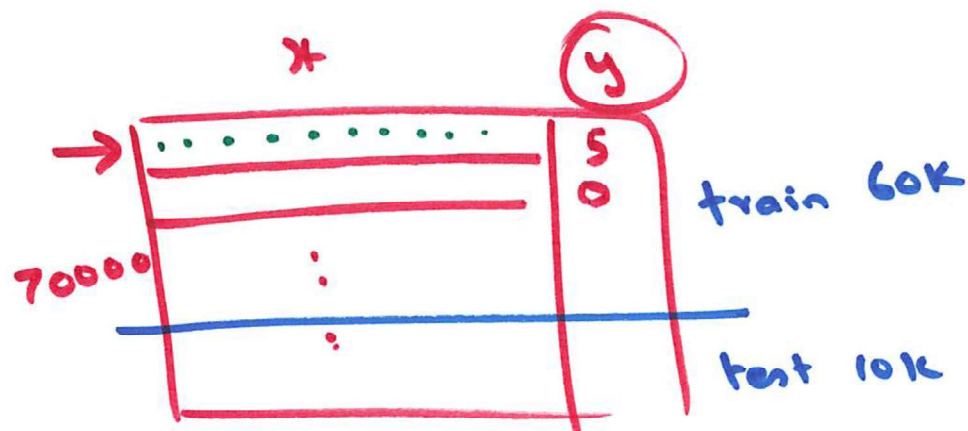
MNIST database

70000

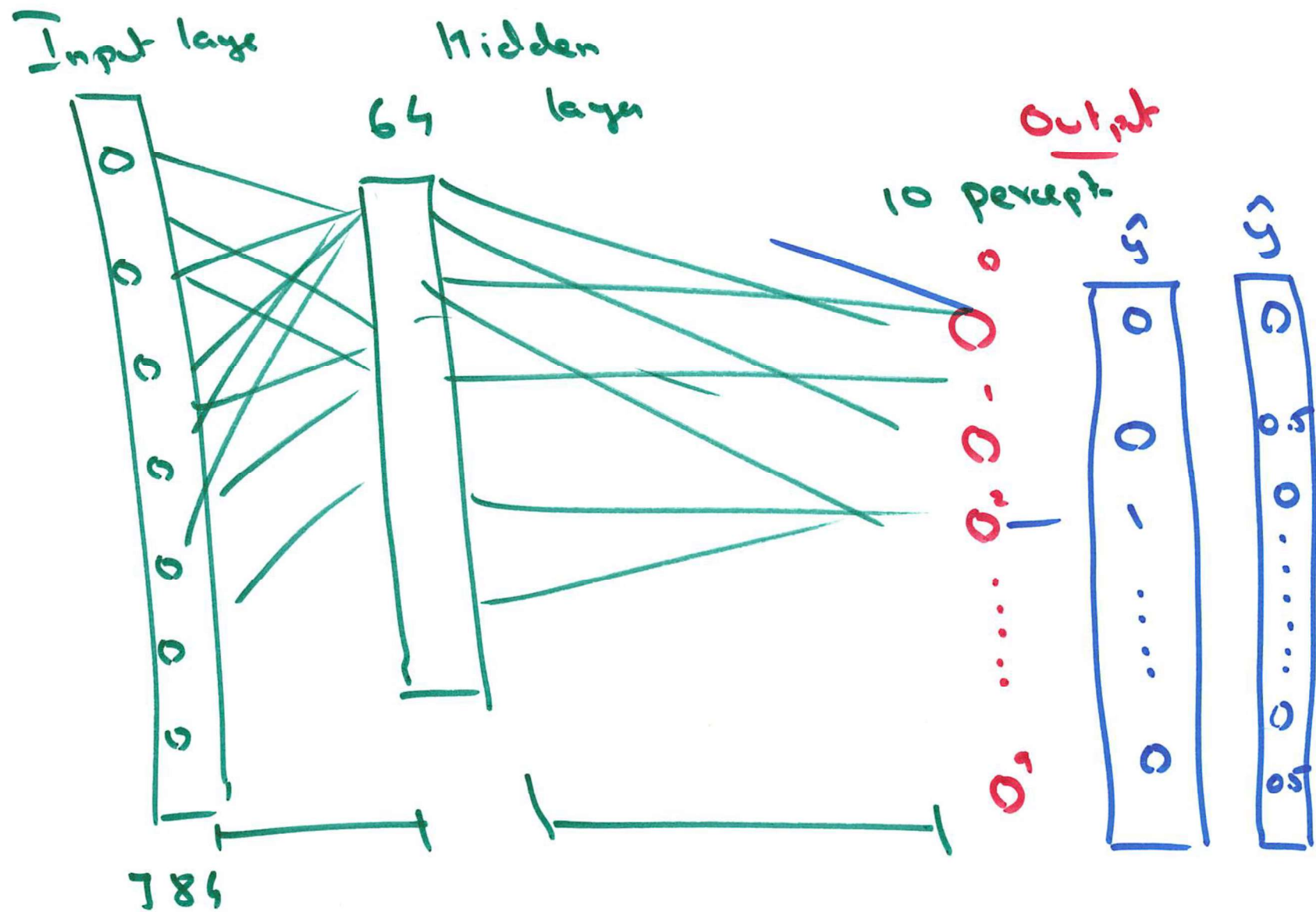
*



$28 \times 28 = 784$

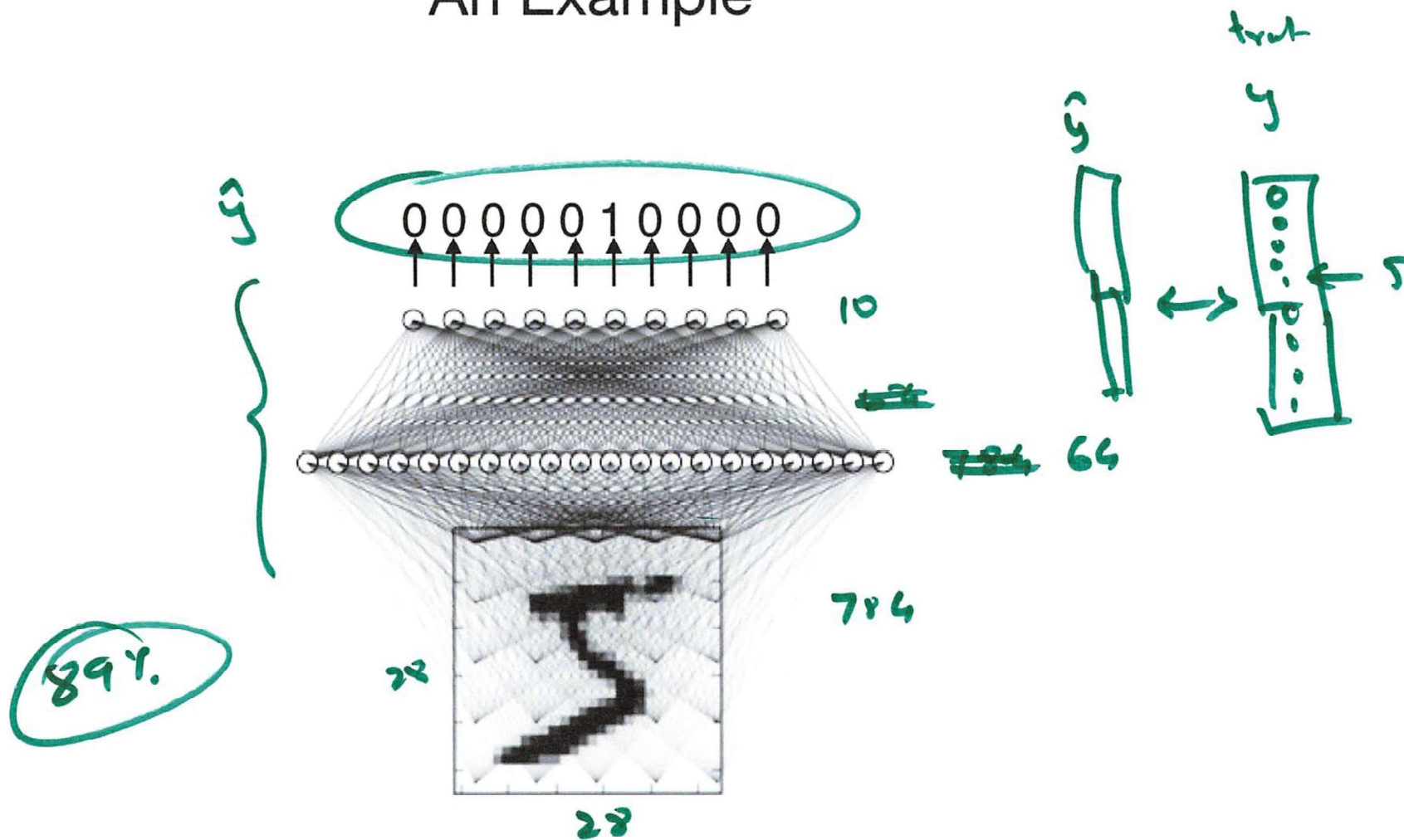


1



$$(784 \times 64 + 64) + ((64 \times 10) + 10) = \underline{\underline{50890}}$$

An Example



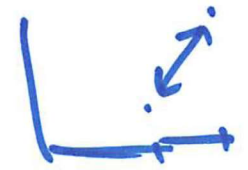
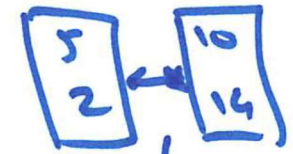
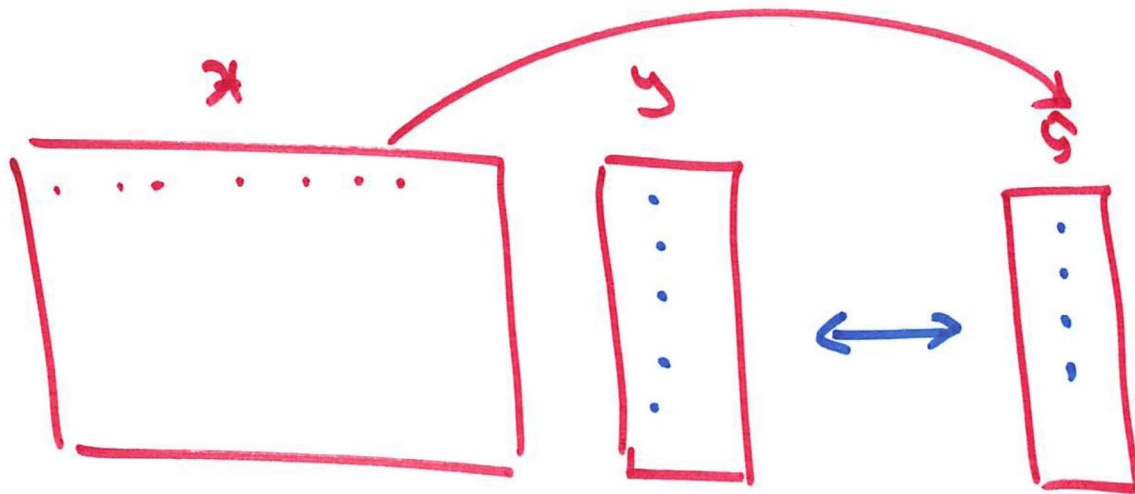
Activation Func

$\sigma()$

$\tau()$

$$z_i = \sigma(\sum w_i x_i + b)$$

$$\tau \left(\tau \left(\tau \left(\dots \right) \right) \right)$$



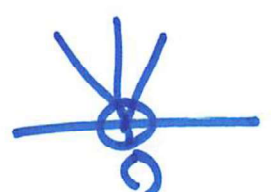
$$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

how far are $\hat{y}$ + y ?

Loss function

$$\frac{1}{2} \sum_{i=1}^n (\hat{y}^i - y^i)^2$$

$$\sum_{i=1}^n |\hat{y} - y|$$

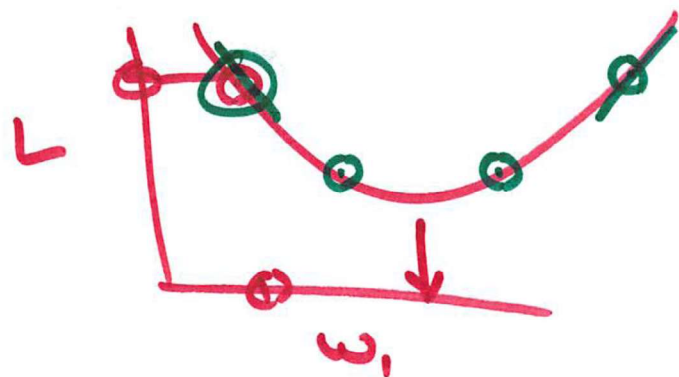


Training

$$\min \sum (\hat{y}^i - y^i)^2 \quad \text{by changing } w_1, w_2, w_3, b$$



$$\min \sum_i \left((w_1 x_i + w_2 x_i + w_3 x_i + b) - y_i \right)^2 = L(w, b) \quad \text{by changing } w_i, b$$



if Slope ~~is~~ ^{Neg}
if Slope ~~is~~ ^{Pos}

then
more $w_1 \rightarrow$
more $w_1 \leftarrow$

$$w_i^{\text{new}} = w_i^{\text{old}} - \eta (\text{Slope of } w_i)$$

↑
learning rate

Gradient Descent

